

ESP32 Vacuum Cleaning Robot Project Report

This project demonstrates the design and implementation of an autonomous and Bluetooth-controlled vacuum cleaning robot using ESP32. The robot uses ultrasonic sensors and IR sensors for obstacle detection and navigation. A blower fan is used for dust suction, and collected dust is stored inside a compact dust box.

1. Objectives

- Design a compact robotic vacuum cleaner.
- Implement autonomous obstacle avoidance.
- Enable Bluetooth manual control.
- Collect dust using suction mechanism.
- Use ESP32 for intelligent control.

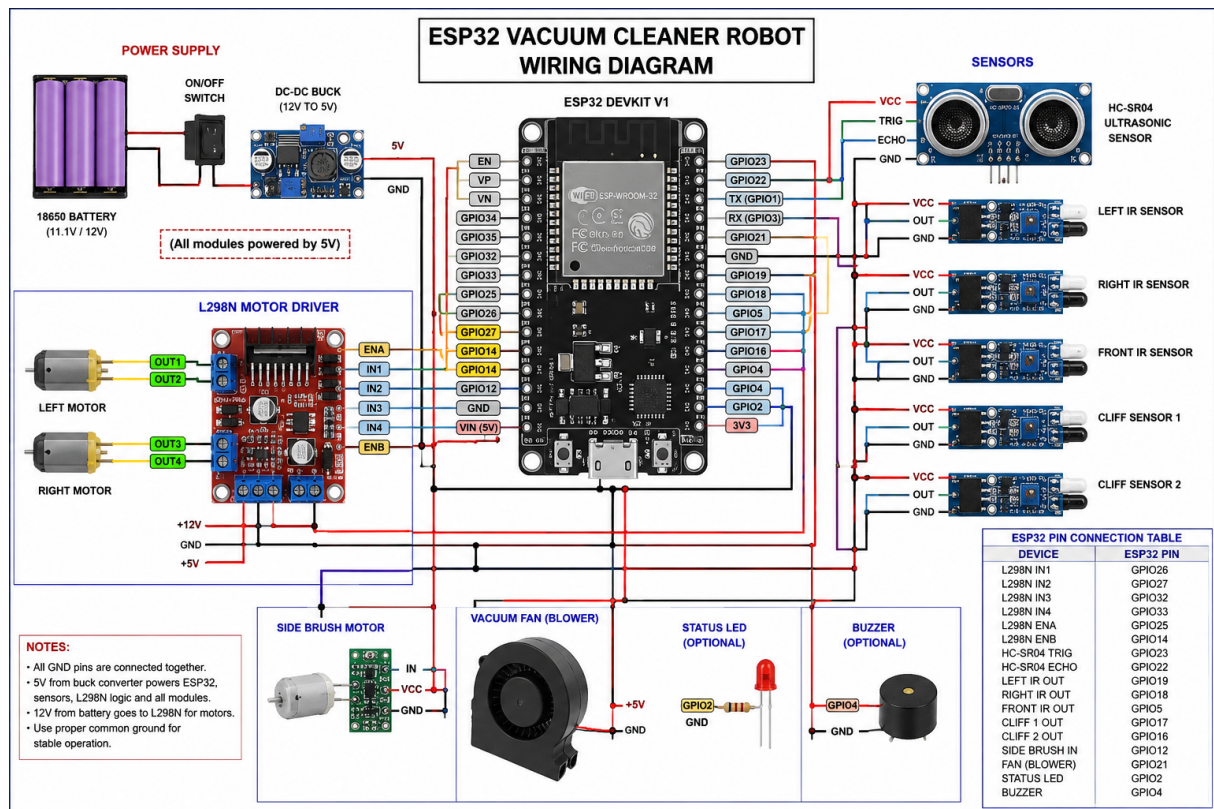
2. Components Used

Component	Quantity
ESP32 Dev Module	1
L298N Motor Driver	1
HC-SR04 Ultrasonic Sensors	2
IR Sensors	2
BO Motors with Wheels	2
18650 Batteries	2
LM2596 Buck Converter	1
Centrifugal Blower Fan	1
Switch	1
Cardboard Chassis	1

3. Working Principle

The ESP32 controls the movement of the robot using the L298N motor driver. Ultrasonic sensors detect obstacles in front and side directions. IR sensors help in edge and close-range detection. The centrifugal blower creates suction to collect dust into the dust box. The robot can operate autonomously or through Bluetooth mobile control.

4. Wiring Diagram



5. Features

- Autonomous obstacle avoidance
- Bluetooth mobile control
- Dust suction system
- Rechargeable battery system
- Compact circular chassis
- Low-cost DIY design

6. Applications

This robot can be used for small-room cleaning, educational robotics projects, DIY automation learning, and smart home prototypes.

7. Conclusion

The ESP32 Vacuum Cleaning Robot is a low-cost smart cleaning solution developed using easily available components. The project demonstrates embedded systems, automation, motor control, and IoT concepts in a practical way.

Prepared using ESP32, Arduino IDE, and open-source electronics components.